

# CASA Renovations: Examining Social Responses to an Anthropomorphic Media Representative That Is Separate From the Core Technology Being Represented

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## Abstract

To extend our understanding of CASA and account for changes in technological complexity and configuration, this research asks: Do increasing levels of social cues in a media representative that is separate from the core technology it represents lead to increased perceived anthropomorphism of the core technology? And does such perceived anthropomorphism predict positive treatment of the core technology? We conducted two complementary studies observing e-scooter use and treatment, an online experiment focused on responses to video stimuli ( $N = 280$ ) and a field experiment focused on actual user behavior ( $N = 39,204$ ). Findings suggest that, indeed, social cues in a media representative (i.e., an e-scooter rental mobile application) of a separate core technology (i.e., the e-scooter itself) leads to positive attitudes about the core technology, potentially because of increased perceived anthropomorphism.

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## Introduction

The Computers-Are-Social-Actors (CASA) framework suggests that people mindlessly treat technologies (e.g., media) in a social manner despite knowing technologies are not social beings (Nass & Moon, 2000; Reeves & Nass, 1996). Through over 30 years of research, our understanding of CASA has become more nuanced, recognizing that the phenomenon is not binary (i.e., responses are either social or nonsocial, Gong, 2008). The ways individuals respond socially to technologies have changed as new cultural norms of human-media social scripts emerge and technologies utilize increasingly complex and sometimes anthropomorphic interfaces (Gambino et al., 2020). While most CASA research has focused on interactions with screen-based media (Carolus et al., 2018; Hoffmann et al., 2009) or automobiles (e.g., K. J. Lee et al., 2014; S. Lee et al., 2019; Nass et al., 2005), the present research extends our understanding of CASA in the novel context of communication with a media interface (i.e., a mobile phone application) that is physically separate from but serves as a representative of the core technology (i.e., rentable electric scooters).

This research takes a variable-centered approach (Nass & Mason, 1990), meaning we focus on a unique technological affordance that is applicable to multiple technologies. The ability for a media agent to represent a separate core technology is exemplified within the specific platform of focus here, rentable e-scooters, but is applicable across a wide range of current and future technologies that share this affordance. For example, people increasingly interact with media representatives of separate technologies, such as mobile applications to control smart home devices (e.g., turn on lights), order rideshares, or to control automobile functionalities (e.g., unlock the doors, climate control). Media representatives also influence how people interact with services and organizations, such as mobile applications to check in for doctor's office visits, to book travel, and to order food in restaurants. We imagine that in such cases, the media representative might serve as a sort of avatar for the core technology. In this way, this study extends human-machine communication research to examine how social cues portrayed by the media representative shape user perceptions of the core technology. Hence, the present research addresses two overarching research questions. First, *do increasing levels of social cues in a media representative that is separate from the core technology it represents lead to increased perceived anthropomorphism of the core technology?* And second, *does such perceived anthropomorphism predict positive treatment of the core technology?*

We addressed these questions in two complementary studies. The first was an online experiment in which participants viewed videos with different levels of social cues embedded in the e-scooter rental process. The second was a field experiment in which anthropomorphic messages were displayed during the actual scooter rental process for a subset of users. Both studies examined how these different interfaces influenced usage (i.e., intent in Study 1, actual rental behavior in Study 2) and responsible attitudes about e-scooters (i.e., norms about vandalism in Study 1, responsible parking behavior in Study 2). Results extend our theoretical understanding of the technological configurations to which CASA

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can apply, suggesting practical implications regarding anthropomorphic media interfaces that represent other core technologies, such as e-scooters.

## CASA Renovations

The CASA framework has been extensively studied in a variety of research domains for nearly 30 years (Fortunati & Edwards, 2021) and our understanding of CASA is changing (e.g., Cambre & Kulkarni, 2019; Castro-Alonso et al., 2021; A. Kim et al., 2019; Tudor Car et al., 2020). For instance, studies suggest the phenomenon is not a binary in which people either do or do not respond socially to technologies with social cues, but instead that an increasing level of social cuing leads to a coinciding likelihood of social responses to a technology (Gong, 2008). Importantly, people's technology-related experiences have also changed drastically in the past 30 years (Gambino et al., 2020). For example, a significant increase in the use of communication technologies like smartphones (Pew Research Center, 2021) has led to the development of human-machine social scripts that differ from the scripts followed during human-human interactions (Edwards et al., 2019; Gambino et al., 2020). Further, advancements in technologies like artificial intelligence, machine learning, speakers, sensors, and motors have facilitated more humanlike technologies (Gambino et al., 2020). Together, a cyclical dynamic seems to emerge: as technologies are imbued with more social cues, people respond more socially to them. However, as people become familiar with these techno-social cues, they respond less socially until the next cycle of technological advancement in social cueing potential.

To elucidate the implications of this advancement, we examine how individuals respond to a core technology when their interactions are mediated through a separate digital interface. This configuration arises from the so-called Internet of Things (IoT), which integrates physical objects with internet connectivity to presumably enhance human interaction with these devices (Kaushik & Bagga, 2020). Among its many sociotechnical implications, the IoT enables verbal and nonverbal communication with any internet-enabled object via a “mediated representative.” This representative can reside on an entirely separate platform from the core technology—such as a mobile phone, smart speaker, or television—as long as the representative has the necessary input (e.g., keypad, microphone) and output (e.g., screen, speaker) components for social interaction with users. Modern examples of this configuration include using a smartphone or smart speaker to operate a robot vacuum or home thermostat, as well as renting an electric scooter through a mobile application. This investigation focuses on the latter scenario because operating the mobile application is distinct from operating the scooter itself.

## The CASA Continuum, From Utilitarian to Social

Individuals use technologies as utilitarian tools to fulfill their functional needs (de Graaf et al., 2018; Mansell & Silverstone, 1996; Rogers, 1983). A large body of literature on technology acceptance (Davis, 1989; Marangunić & Granić, 2015) suggests that adoption of technology is largely dependent on its perceived usefulness, which differs by individual characteristics—like gender (Anderson et al., 2008; T. Kim et al., 2023; Young, 2000), race (T. Kim et al., 2023), age (Araujo et al., 2020; T. Kim et al., 2023), or education level (Araujo

et al., 2020; T. Kim et al., 2023)—and technological characteristics, like how well the technology achieves user goals.

Technologies may also provide social and relational benefits in addition to their utility (de Graaf et al., 2018; Xu & Li, 2022), even though they are not human. The CASA framework highlights that people often mindlessly perceive technologies as social actors, attributing to them humanlike characteristics such as gender, social roles (e.g., teammates, specialists, and flatterers), and personalities (Nass et al., 1994, 1995, 1996). Nass and Moon (2000) proposed that we tend to mindlessly respond to technologies in social ways because the human brain has evolved for interpersonal communication to unconsciously react socially to social cues (Nass & Moon, 2000). For example, people tend to respond favorably to flattering computers, even when they know such flattery is meaningless (Moon, 2000), illustrating the human tendency to reflexively respond positively to praise and affirming feedback.

The CASA paradigm suggests that people interact with computers socially not because they intentionally anthropomorphize them, but rather because they effortlessly and unintentionally respond to anthropomorphic cues. Nass and Moon (2000) stated that a technology should present enough social cues for users to exhibit social responses toward the technology. Even though they did not clarify what “enough” social cues are, mindlessness requires a certain number of social cues to be activated.

Building on this concept, research indicates that anthropomorphism can occur both mindlessly and mindfully (Y. Kim & Sundar, 2012). Specifically, mindful and mindless anthropomorphism lie along a spectrum influenced by the number of social cues presented (Lombard & Xu, 2021). As the degree of perceived anthropomorphism increases, individuals also tend to form more positive social judgments (Gong, 2008). The proposed medium-as-social-actor (MASA) paradigm suggests that there is a hierarchy of social cues that can be embedded into a medium to induce anthropomorphism, where only a certain quantity is necessary to facilitate *mindful* anthropomorphism. However, additional quantity and higher quality cues are required to shift users to more *mindless* anthropomorphism (Lombard & Xu, 2021). This approach implies a linear relationship between social cues and perceived anthropomorphism, which research suggests might not be the case for high quality social cues due to the uncanny valley effect (Mori, 1970; Mori et al., 2012), but this issue is not yet relevant in the current research given the focus on relatively low fidelity social cues. Hence, taken together, social cues should be positively related to the perception of technologies as humanlike, which should relate to positive perceptions of that technology.

The current study extends this line of reasoning by examining how the extent of social cues embedded in the mediated representation of a core technology influences the perceptions of the represented core technology. We posit that because the mediated representative is intrinsically linked with the core technology, the more social cues in the mediated representative, the more the core technology will be perceived as humanlike:

**H1:** The more humanlike features (e.g., voice and/or name) a mediated representative of a technology displays, the more users will perceive the represented technology to be anthropomorphic.

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Building on the previous logic, we posit that such humanlike features in a mediated representative will lead to more positive attitudes about the core technology, with this effect mediated by the extent of perceived human-likeness. For the present work, we observe positive attitude outcomes across two categories: endorsement of responsible social norms and usage intent. Regarding the former, we consider *antisocial* or otherwise irresponsible treatment of technology, such as cases of robot abuse (Bartneck et al., 2005; Nomura et al., 2015), as the opposite of positive treatment. Endorsement of responsible social norms—meaning guidelines for behavior in society (Shulman et al., 2017) that can be both proscriptive (i.e., what we should do) or prescriptive (i.e., what we should not do; Janoff-Bulman et al., 2009; O'Neill, 2017)—opposed to such antisocial treatment (i.e., a prescriptive norm) would thus represent a positive attitude about the core technology. Regarding usage intent, we posit that the perceived human-likeness of the mediated representative will lead to more positive attitudes about the utility of the core technology due to a halo effect, which is the over-attribution of positive attitudes about someone because of an overreliance on a single characteristic, such as attractiveness (Batres & Shiramizu, 2022). In this case, we expect that increases in perception of the mediated representative as humanlike—somewhat akin to social attractiveness—will lead to increased perception of utility and, thus, usage intent. We articulate these expectations about the endorsement of responsible social norms and usage intent as follows:

**H2:** Perceived anthropomorphism will mediate the effect of the mediated representative technology's human likeness on the endorsement of social norms for the represented technology.

**H3:** Perceived anthropomorphism will mediate the effect of the mediated representative technology's human likeness on the technology usage intent of the represented technology.

To expand on our expectation regarding usage intent, we turn to the technology acceptance model (TAM), arguably the most widely adopted theoretical model of technology adoption, which posits that perceived usefulness (PU) and ease of use (EU) are key predictors of technology usage intent (Davis, 1989; Marangunić & Granić, 2015). In the current research, we are interested in PU and EU for both the mediated representative and the represented core technology. Further, PU tends to be a stronger and more consistent predictor of use intent than EU—though EU often predicts PU—according to a meta-analysis of TAM research (King & He, 2006). Hence, instead of proposing hypotheses, we pose two open-ended research questions.

**RQ1a.** Do perceived ease of use and/or perceived usefulness of the *mediated representative* act as additional serial mediators in the relationship proposed in H3?

**RQ1b.** Do perceived ease of use and/or perceived usefulness of the *represented technology* act as additional serial mediators in the relationship proposed in H3?

Finally, although TAM focuses on usage intent as the outcome, we are interested in whether PU and EU predict the endorsement of social norms, which is also a positive

perception of the represented technology. Hence, we pose these two open-ended research questions.

**RQ2a.** Do perceived ease of use and/or perceived usefulness of the *mediated representative* act as additional serial mediators in the relationship proposed in H2?

**RQ2b.** Do perceived ease of use and/or perceived usefulness of the *represented technology* act as additional serial mediators in the relationship proposed in H2?

## Study 1: Online Experiment

### Context: E-scooters

Electric scooters (e-scooters), a *micro-mobility* technology, are increasingly used for local transportation (Dozza et al., 2022; Møller et al., 2020). *Rentable* e-scooters—an environmentally friendly alternative to ridesharing in cars (Choron & Sakran, 2019; Dozza et al., 2022; Kobayashi et al., 2019)—offer a novel yet appropriate context for research on interactions with mediated representatives of core technologies because they usually require simple interactions with mobile applications in order to unlock, ride, and return the device (Hawa et al., 2021).

Perceptions of how e-scooters should be treated offer a theoretically appropriate and socially meaningful measure of adherence to social norms. Norms of e-scooter treatment are codified in the terms of service agreement or even laws, such as riding safely and within speed limits (Choron & Sakran, 2019). Still, these norms are often defied, hindering public safety (Gössling, 2020; Zakhem & Smith-Colin, 2021). Public sentiment about e-scooters can be quite negative, leading some to commit acts of vandalism (Moreau et al., 2020), including burning e-scooters, throwing them into bodies of water, or improperly parking them (see Figure 1, Appendix B).

Negative emotional responses to e-scooter vandalism reflect a prescriptive social norm that others should not mistreat e-scooters, just as others should not mistreat people according to many ethical principles (e.g., the “golden rule”). We consider this a social norm based on expectations of behavior with members of a community.

### Study 1: Methods

Participants from a large Midwestern university received a small amount of course credit to complete an IRB-approved online experiment, which involved viewing an approximately 2-minute video depicting a process of renting and riding an e-scooter, followed by a questionnaire on perceptions of the video. Participants were randomly assigned to view one video out of five possible conditions based on the Qualtrics software’s random assignment tool. There were three conditions of the scooter messaging style: text only (no anthropomorphic cues), utilitarian voice (low anthropomorphic cues), and social voice (highest anthropomorphic cues). There were two voice conditions—female-sounding and male-sounding—included as a control variable in the study because of previous CASA research suggesting that voice gender makes a difference in perceptions of transportation technologies (e.g., S. Lee et al., 2019). Because the scooter gender was only conveyed in the

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voice conditions, the conditions were not fully crossed but instead followed a 1 (text only) + 2 (voice messaging style)  $\times$  2 (voice gender) design (See Figure 5). Hence, comparisons between conditions were constrained to test hypotheses as appropriate.

The survey was accessed 284 times, of which four participants entered the survey twice. For three of these four, their original entries were 95% incomplete because participants selected “I am not at all familiar with e-scooter rentals” at the start of the survey and were automatically sent to the end of the survey. These three returned to the survey, presumably because they recognized this filtering mechanism, and then selected, “I am somewhat familiar with e-scooter rentals. I know they exist, but I have never used one myself.” These completions were included in the analysis. The other participant originally completed 44% of the survey and then exited the survey before viewing the video, possibly because they were interrupted, and then returned to complete the survey. This person’s data was included in the analysis.

The final sample size was  $N = 280$ , including 164 women and 114 men, one nonbinary person, and one not reporting gender. These latter two participants were included in the descriptive statistics and correlational analyses but excluded from the tests that used participant gender as a predictor variable, namely, the analyses of variance (ANOVAs) and mediation analyses ( $n = 278$ ). The majority of participants self-identified as White ( $n = 201$ ), 41 as Asian, 20 as Black or African American, four as Arab or Middle Eastern, four as other, and 10 as multi-racial. The average age was 20.36 years old ( $SD = 1.36$ ). A total of 56 participants were assigned to the text-communication-only control, 113 to the utilitarian condition (57 in female and 56 in male e-scooter voice conditions), and 109 to the humanlike condition (54 in female and 55 in male e-scooter voice conditions).

## Materials

Our research team iteratively designed the experimental stimuli script and messaging (see Appendix C for the full script), which consisted of seven 2-minute first-person point-of-view videos depicting someone renting, riding, and parking an e-scooter managed by the brand, SPIN. The SPIN brand was chosen because of its widespread familiarity and use at the university where this research was conducted, and due to the SPIN brand’s role as a supporting partner and consultant on this research. The final videos were shot using a chest-mounted GoPro HERO8 camera, then edited and mixed by a member of the research team with professional-level video and audio editing abilities using *Adobe Premiere Pro*, *After Effects*, and *Mocha*.

Each video included the same base footage—without any e-scooter messaging—depicting an augmented reality experience where the user navigates through notifications, maps, and the SPIN e-scooter mobile application. The storyline portrays the user exiting a building, noticing they are late to a dinner reservation, finding and renting an e-scooter, riding for an extended period, encountering a “slow zone” where speed is limited, and parking and confirming via the application (see Figure 2, Appendix A). This base footage, with only simple text-based messages portrayed from the application, served as the control (no anthropomorphic cues). Scooter voice messages for each of the other respective conditions were later edited onto this sequence.

The e-scooter messaging-style conditions included on-screen text notifications that were motion-tracked to the e-scooter (e.g., moved with the e-scooter as if it existed as an extension of it) alongside text-to-speech male and female voiceovers which were generated using *Adobe Audition*. The specific notifications were designed to highlight scooter-relevant information, such as the availability of an e-scooter for rental (see Appendix C for all messages). Messages were iteratively designed by the research team to depict the same basic information with different levels of human-likeness depending on the condition, e.g., “The e-scooter is here, click to rent *the* e-scooter” (utilitarian condition: low anthropomorphic cues) or “Hi, *Spinnigale* here, I’m close by!” (social condition: high anthropomorphic cues). To portray the e-scooters as humanlike, they were given proper gendered names, namely “Spinnigan” and “Spinnigale.” For examples of e-scooter messaging across conditions, see Figures 3 and 4 (Appendix A).

## Measures

Specific measures for these constructs were based on responses to 1-to-5 Likert-type scales of agreement, averaged across scale items. Reliability values were acceptable for all constructs ( $\alpha > .70$ ). Descriptive statistics and inter-construct correlations can be found in Table 1 and Table 2, respectively (Appendix B).

**Perceived anthropomorphism** was measured using the six-item perceived anthropomorphism scale developed by Moussawi and Koufaris (2019). Participants ranked the extent to which they felt the e-scooter in the video “*is able to speak like a human*” and if the e-scooter can be “*happy*,” “*friendly*,” “*respectful*,” “*funny*,” and “*caring*.”

**Social norms of e-scooter treatment** were measured through the anticipated negative emotion scale (Rosenthal & Ho, 2020; Zhuang & Bresnahan, 2017) adapted for this context, measuring participant normative responses to hypothetical scenarios of e-scooter vandalism. Participants were asked to “*imagine if the e-scooters from the video were in your community*” or “*neighborhood*,” and respond to six items indicating whether they would be embarrassed or ashamed if these e-scooters were vandalized, damaged, or not safely used.

**Perceived usefulness** (PU) and **ease of use** (EU) were each measured using three-item scales developed by Venkatesh (2000). Responses measured feelings of usefulness and ease of use toward both the representative technology (SPIN mobile application) and the represented core technology (e-scooter).

**Usage intent** for the e-scooter was measured using a four-item scale developed by Venkatesh et al. (2012). Participants were asked to indicate to what extent they “*intend to use*,” “*will try to use*,” “*expect [to] use*,” and “*plan to use [the e-scooter] in the future*.”

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## Study 1: Results

To test H1 (the more humanlike features in a mediated representative, the more users will perceive the represented technology to be anthropomorphic), we conducted a two-way ANOVA with perceived anthropomorphism as the dependent variable, scooter messaging style (utilitarian vs. social) and scooter voice gender (male vs. female) as independent variables, and participant gender as a covariate. This model did not include the text-only condition in order to be able to examine the  $2 \times 2$  interaction between scooter messaging style and voice gender (see Table 3, Appendix B). Levene's test showed that homogeneity of variance assumptions were met,  $F(3, 218) = 0.88, p = .45$ . Results showed that perceived anthropomorphism was significantly influenced by scooter messaging style,  $F(1, 217) = 13.83, p < .001, \eta_p^2 = .06$ , but not scooter voice gender ( $p = .16$ ), participant's gender ( $p = .72$ ), or the interactions between messaging-style and voice gender ( $p = .55$ ).

We then conducted a one-way ANOVA with perceived anthropomorphism as the dependent variable and all three messaging styles (text-only vs. utilitarian vs. social) as independent variables. Levene's test indicated a violation of homogeneity of variance,  $F(2, 275) = 7.71, p < .001$ . Following a common approach to addressing such a violation (Welch, 1951), we conducted a Welch's ANOVA test, which revealed statistically significant differences in perceived anthropomorphism across messaging styles,  $F(2, 160) = 27.4, p < .001$ .

Games-Howell post hoc tests showed perceived anthropomorphism was significantly higher in the utilitarian voice ( $M = 2.02, SE = .09; p < .001$ ) and social voice ( $M = 2.53, SE = .10; p < .001$ ) conditions compared to the text-only condition ( $M = 1.47, SE = .13$ ). Additionally, perceived anthropomorphism was significantly higher in the social voice compared to the utilitarian voice condition ( $p < .001$ ). Thus, H1 was supported.

To test H2 (perceived anthropomorphism will mediate the effect of the mediated representative technology's human likeness on the **endorsement of social norms**), we conducted a mediation analysis using PROCESS for R v4.3.1 (Model 4; 10000 bootstrap samples; Hayes, 2022), with e-scooter communication-style (social vs. utilitarian, with utilitarian as the reference group) as the independent variable, social norms endorsement as the dependent variable, perceived anthropomorphism as the mediator, and e-scooter voice gender and participants' gender (female vs. male, with male as the reference group for both) as covariates.<sup>1</sup> The partially standardized relative indirect effect from the social e-scooter messaging style on social norm endorsement when mediated by perceived anthropomorphism of the e-scooter was statistically significant ( $\beta = 0.13, 95\% \text{ C.I.}[0.05, 0.24]$ ), while the direct effect of e-scooter messaging style on social norms endorsement was not found to be significant ( $\beta = -.05, 95\% \text{ C.I.}[-0.31, 0.20], p = .67$ ). This indicates full mediation, supporting H2.

We conducted a similar mediation analysis to predict usage intent, testing H3 (perceived anthropomorphism will mediate the effect of the mediated representative technology's human likeness on **usage intent**). The partially standardized relative indirect effect of e-scooter voice style on usage intent through perceived anthropomorphism was found

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1. These mediation analyses do not include the text-only condition in order to retain voice gender as a covariate.

to be statistically significant ( $\beta = 0.09$ , 95% C.I.[0.02, 0.19]), whereas the direct effect from e-scooter voice style on usage intent was not found to be significant ( $\beta = .13$ , 95% C.I. [-0.14, 0.40],  $p = .35$ ). As such, this also suggests full mediation, supporting H3.

We conducted four additional mediation analyses using PROCESS (Model 6; 10000 bootstrap samples) to address RQ1a–RQ2b (do PU and/or EU of the mediated representative or of the represented technology act as additional serial mediators in the relationships proposed in H2 and H3?), with e-scooter communication-style (social vs. utilitarian, with utilitarian as the reference group) as the independent variable, usage intent (RQ1a&b) or social norms endorsement (RQ2a&b) as the dependent variable, perceived anthropomorphism as the mediator, EU and PU of the e-scooter mobile app (RQ1a & RQ2a) or of the e-scooter itself (RQ1b & RQ2b) as additional serial mediators (in EU  $\rightarrow$  PU order), and e-scooter voice gender and participants' gender (female vs. male, with male as the reference group for both) as covariates. The full serial mediation model through **e-scooter mobile app perceptions** (*communication style  $\rightarrow$  perceived anthropomorphism  $\rightarrow$  EU e-scooter mobile app  $\rightarrow$  PU e-scooter mobile app  $\rightarrow$  usage intent or norm endorsement*) was statistically significant for both usage intent ( $\beta = 0.02$ , 95% C.I.[0.001, 0.037]) and social norms endorsement ( $\beta = 0.01$ , 95% C.I.[0.000, 0.018]). The full serial mediation model through **e-scooter perceptions** (*communication style  $\rightarrow$  perceived anthropomorphism  $\rightarrow$  EU e-scooter  $\rightarrow$  PU e-scooter  $\rightarrow$  usage intent or norm endorsement*) was not significant. However, when removing e-scooter EU as a serial mediator (i.e., *communication  $\rightarrow$  anthropomorphism  $\rightarrow$  e-scooter PU  $\rightarrow$  intent/ norms*), the model became significant.

Providing further clarity in response to our RQs, the standardized indirect effects were significant for all paths that did not include PU of both the e-scooter itself or the e-scooter mobile app. The standardized indirect effect of e-scooter voice style on **usage intent** through perceived anthropomorphism and **mobile application PU** was 0.08 (95% C.I.[0.030, 0.138]), while through perceived anthropomorphism and **e-scooter PU** was 0.04 (95% C.I.[0.011, 0.079]). The standardized indirect effect of e-scooter voice style on **social norm endorsement** through perceived anthropomorphism and **mobile application PU** was 0.04 (95% C.I.[0.007, 0.072]), while through perceived anthropomorphism and **e-scooter PU** was 0.03 (95% C.I.[0.007, 0.054]). Additionally, the standardized indirect effect of e-scooter voice style on social norm endorsement through perceived anthropomorphism was significant in the model controlling for *mobile application PU and EU* ( $\beta = 0.09$ , 95% C.I.[0.022, 0.187]) and also in the model controlling for *e-scooter PU and EU* ( $\beta = 0.10$ , 95% C.I.[0.031, 0.194]). See Figure 6 from the comparison between four different mediation models.

## Study 1: Discussion

Do increasing levels of social cues in a media representative that is separate from the core technology it represents lead to increased perceived anthropomorphism of the core technology, and does such perceived anthropomorphism predict positive treatment of the core technology? Addressing these questions, Study 1's online experiment found that when a mediated representative (i.e., a mobile application) was depicted with more humanlike features (i.e., text only versus utilitarian voice and social voice), participants perceived the represented technology (i.e., the e-scooter) to be more anthropomorphic. This perceived

anthropomorphism was found to mediate the effect of the human-likeness of a technology's media representative (i.e., e-scooter mobile application), along with perceived ease of use and/or perceived usefulness of the media representative and/or the core technology itself on usage intent and social norm endorsement for treatment of the core technology (i.e., e-scooters). We found nuanced differences in these mediation models regarding the PU and EU of the e-scooter mobile application and the e-scooter itself on the usage intent and social norm endorsement outcomes (see Figure 1 for an overview). Across models, communication style consistently influenced perceived anthropomorphism, which influenced perceived usefulness of both the e-scooter and the mobile app, which influenced both usage intent and social norm endorsement. Perceived anthropomorphism only seemed to be associated with perceived ease of use for the mobile app, not the e-scooter itself. Perceived anthropomorphism seemed to relate directly only to social norm endorsement, not usage intent. Together, these findings expand our understanding of CASA to applications in which the mediated representative is separate from the represented core technology, yet social interactions with the former lead to positive—and less anti-social—perceptions of the latter.

The largest limitation of Study 1 is the reliance on filmed experimental stimuli and self-report attitudinal measures about e-scooters. Study 2 addresses this limitation.

## Study 2: Field Experiment

Study 2 tests the same general conceptualization as Study 1, but because Study 2 was conducted as a field experiment focused on actual user behavior, we were not able to examine the psychological mediators of the proposed effects. However, we were able to test the same basic hypothesis as in Study 1.

**H1.** A mediated representative that displays more humanlike features, compared to a less humanlike mediated representative, will elicit more positive treatment of the represented core technology.

## Study 2: Methods

We conducted an IRB-approved study of the scooter rental user data obtained through a partnership with SPIN, an e-scooter rental company. The data was permissible according to the SPIN user terms of service. Our research team had access only to fully anonymized data, and we did not have access to any potentially identifiable user information. The experiment directly manipulated the actual user interface of the SPIN app that users viewed and used. The experiments took place for a 1-year period (09/01/2022–08/31/2023). During this period, participants were randomly assigned by the SPIN product team to either receive the social message condition or the typical interface. In the social message condition, the e-scooters were given an anthropomorphic name, “Spindy,” and used first-person pronouns (I/me). In contrast, in the utilitarian condition, the user interface referred to scooters as the generic term for the technology. For example, opening the app triggered the message, “Hi! Spindy here! I’m close by. Click on the map to see where I am,” in the

social condition, and “Scooters are close by. Click on the map to see where they are,” in the utilitarian condition.

We collected a total of 30,496 trip data from 1,780 users in the social condition and 36,493 trip data from 2,307 users in the utilitarian condition. Data consisted of geological coordinates of a trip’s starting and ending locations and trip duration. We excluded duplicated trip data and error data (e.g., later trip start time than trip end time) used a total of 30,406 trip data from 1,779 users in the social condition and 36,404 data from 2,306 users in the utilitarian condition. First, based on the coordinates, we calculated the expected travel distances considering the actual road systems and the physical speed limit of SPIN e-scooters using the Stadiamap API. We also calculated the expected average speed of each based on the distance and duration and a ratio of proper parking behaviors using a user’s entire trip numbers and the number of trips that parked an e-scooter in the specially designated e-scooter parking locations on or near the campus. To increase the interpretability of the results, we did not include trip data that happened outside of SPIN’s operating area (social condition:  $n = 10,987$ , utilitarian condition:  $n = 14,623$ ) and potential error data, such as trip data having less than 10 meters (around 32 feet) of trip distances or less than 1 minute of trip duration (social condition:  $n = 853$ , utilitarian condition:  $n = 1143$ ) (Zou et al., 2020). After data cleaning, we aggregated all trip data into an individual user level, and we utilized a total of 18,566 trip data from 1,564 users in the social conditions and 20,638 trip data from 1,938 users in the utilitarian condition (see Table 4 for descriptive statistics, Appendix B).

To explore and confirm data validity, we calculated the correlation between the distance and duration, and unexpectedly, we found a negative correlation between them ( $r = -.11$ ). A subsequent scatterplot revealed potential outliers with excessively long durations and short trip distances (Figure 7). Because we were not able to identify whether the outliers were caused by errors or abnormal user behaviors (e.g., forget to end the trip, riding for fun with the same start/endpoints, etc.), we created an additional dataset restricted to trip durations within the range of 3 SDs of the mean (i.e., maximum duration of 20 minutes). In this sample, there was a total of 17,378 trip data from 1,534 users in social conditions and 18,932 trip data from 1,877 users in the utilitarian condition. Distance and duration were positively related ( $r = .24$ ), and the scatter plot did not indicate outliers (Figure 8). We conducted our main analyses on both datasets to bolster our findings’ reliability.

## Study 2: Results

We conducted our analyses first on the full dataset (Figure 9). According to Levene’s test (see Table 5), all the variables, except for the number of trips, did not satisfy the assumption of homogeneity of variance. For those variables, we conducted Welch’s  $t$ -test, a reliable alternative to the Student’s  $t$ -test (Delacre et al., 2022). We also calculated Cohen’s  $d$  for Student’s  $t$ -test (Cohen, 1988) and effect size  $e$  for Welch’s  $t$ -test, suggested by Aoki (2020). We found a significant difference in ride distance [ $t(3421) = -3.51$ ,  $p = .000$ ,  $e = .12$ ], duration [ $t(3422) = -3.48$ ,  $p = .001$ ,  $e = .12$ ], and speed [ $t(3443) = -5.16$ ,  $p = .000$ ,  $e = .18$ ] with participants in the social condition traveling farther ( $M = 0.712$  miles,  $SD = 0.277$ ), with shorter durations ( $M = 11.26$  minutes,  $SD = 11.92$ ), and faster ( $M = 5.95$  mph,  $SD = 2.59$ ) than participants in the utilitarian condition (distance:  $M = 0.68$  miles,  $SD = 0.30$ ;

duration:  $M = 12.79$  minutes,  $SD = 13.55$ ; speed:  $M = 5.48$  mph,  $SD = 4.49$ ). The effect sizes for these differences are small (Aoki, 2020) but still detectable due to the large sample size. We did not find a significant difference between conditions for the average number of trips [ $t(3500) = -1.86$ ,  $p = .063$ ,  $d = .063$ ]. Finally, users who were exposed to the social message condition were significantly more likely to park their e-scooters in designated parking areas [ $M = 52.50\%$ ,  $SD = 35.40\%$ ;  $t(3396) = -3.15$ ,  $p = .002$ ,  $e = .11$ ] compared to those in the control group ( $M = 48.60\%$ ,  $SD = 36.90\%$ ).

We then conducted the same analyses on the restricted dataset (Figure 10). We conducted Welch's  $t$ -test for the speed of trips and proper parking ratio because they did not satisfy the assumption of homogeneity of variance (Table 5). Compared to findings using the entire dataset, the results of statistical analyses were the same. There were no significant differences in the number of trips [ $t(3197) = -1.40$ ,  $p = .16$ ,  $d = .05$ ], but the social condition led to shorter durations [ $t(3142) = -2.88$ ,  $p = .00$ ,  $e = .10$ ] and longer distances [ $t(3197) = -2.27$ ,  $p = .02$ ,  $d = .08$ ] with more proper parking ratio [ $t(3143.572) = -2.42$ ,  $p = .02$ ,  $e = .09$ ].

However, there was no significant difference in the trip duration between the two conditions [ $t(3197) = 0.33$ ,  $p = .74$ ,  $d = .01$ ], which was unsurprising given that this sample was restricted by duration.

## Study 2: Discussion

Study 2 investigated the influence of social message intervention on e-scooter usage patterns among users. The results suggest that compared to people who rented e-scooters using the mobile application's basic messaging, participants who received a social message rode e-scooters for longer distances, shorter durations (in the full sample dataset only), and higher speeds and were more likely to park their e-scooters in designated parking areas. The lack of a significant difference found in the number of trips might suggest support for the null hypothesis, given the large sample size in these tests, but this interpretation should be treated with caution because we did not conduct equivalence testing. Still, the behaviors we did find to be associated with social messaging might reflect more positive orientations toward e-scooters, which is consistent with the findings from Study 1.

The behavioral differences we found might reflect positive e-scooter treatment for a variety of reasons. Proper parking behaviors are clearly indicative of respect for community norms, as examined in Study 1. Trip distance, duration, and speed are more difficult to interpret. They may reflect a greater enjoyment of e-scooter use. Riders in the social messaging condition may have chosen to ride for longer distances—instead of stopping to park the scooter at a closer location—and may have chosen to do so because the social messaging experience oriented them toward the enjoyment they felt while riding. Similarly, riding faster might reflect more enjoyment in this context where the maximum speed was limited technologically by the e-scooter to a safe range, and the mean speed was still relatively low (under 5 mph). These interpretations are speculative, of course. Future research could use other methodologies, such as experiments with smaller samples, to confirm whether social (compared to utilitarian) messaging causes further and faster e-scooter riding and whether rider enjoyment mediates this relationship.

Although the effect sizes were relatively small (and smaller in the restricted sample dataset), this study provides more ecologically valid evidence that humanlike features imbued into a mediated representative (i.e., the e-scooter rental application) can elicit more positive behaviors related to the represented core technology (i.e., the e-scooter itself).

## Overall Discussion

The present research asks the question: Do increasing levels of social cues in a media representative (i.e., an e-scooter rental mobile application) that is separate from the core technology it represents (i.e., the e-scooter) lead to increased perceived anthropomorphism of the core technology and does such perceived anthropomorphism predict positive treatment of the core technology? Findings from Study 1, an online experiment in which participants viewed videos of the e-scooter rental process, suggest that more humanlike features (i.e., text only versus utilitarian voice and social voice) increased perceived anthropomorphism and perceived usefulness, which mediated an effect on positive treatment of the technology (i.e., endorsement of social norms and usage intent). Findings from Study 2 extended these insights in a field experiment focused on actual user behavior, suggesting that e-scooter users who received social messages from an e-scooter persona rode e-scooters for longer distances and shorter durations and exhibited more responsible parking behaviors. Together, these findings suggest that, indeed, social cues in a media representative of a separate core technology lead to more positive attitudes about the core technology because of increased perceived anthropomorphism.

Although the media equation is theorized to be a mindless effect that occurs regardless of conscious anthropomorphism of the technology (Nass & Moon, 2000), these studies found that the amount of imbued and, thus, perceived anthropomorphism influences social responses to the technology. This is consistent with previous studies' findings that the amount of agent anthropomorphism is positively related to perceptions of agent trustworthiness, competence, and self-similarity (Gong, 2008; Waytz et al., 2014). Taken together, the CASA paradigm should consider changes in our technological and cultural landscape (Gambino et al., 2020), such as the potential to design technologies in varying degrees of anthropomorphism.

Further, imbuing media representatives with anthropomorphic cues may yield benefits for the technologies being represented. This represents a potentially fruitful direction for human-machine communication research as such media representatives are integrated into a wider range of technologies. For example, people who use smart speakers to interact with home devices likely perceive the devices in certain ways depending on their interactions with the smart speakers, despite the two being entirely distinct (albeit networked) technologies. This conceptual distinction between the representative and represented technology offers grounds for novel theoretical extension of CASA research and other approaches to human-machine communication. Such work should also consider contexts in which the media representative serves as a gatekeeper to interactions with services or organizations that are not usually considered technologies (e.g., mobile applications for account management, e-commerce, etc.). In this way, especially when imbued with anthropomorphic cues, the media representative may serve as a sort of avatar for the service or

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organization it is representing. As natural language interfaces are increasingly powered by artificial intelligence and integrated into everyday technologies, we might expect that any computational interface could be designed to act as an anthropomorphic representative. Future HMC research should extend the discussion of CASA to examine how such avatars of technologies, organizations, and services influence users.

## Limitations

Study 1's most notable limitation was the use of video-based stimuli rather than actual technology use. Study 2 addressed this limitation using a field experimental design but was not able to assess both behavior and user attitudes simultaneously. Thus, differences found relating to trip distance, duration, and speed are difficult to interpret with certainty. We have speculated that rider enjoyment is a potential causal mechanism here, given that enjoyment is a known factor in technology acceptance (Davis, 1989; Marangunić & Granić, 2015), but future research should utilize designs that directly test for this path from social responses to enjoyment to technology use.

As another limitation, there was a potential confound in Study 2 in the user experience of the application besides the communication with the agent as we designed it. Users in the treatment condition may have experienced a novelty effect, especially if they were very familiar with the mobile app prior to the study, which may have influenced their behavior in ways unrelated to the social nature of communication.

## Conclusion

This research provides evidence that adding anthropomorphic cues to a media representative of core technology, such as e-scooters, facilitates positive treatment of that technology, such as stronger endorsement of the social norms related to interactions with the technology (e.g., anti-vandalism attitudes), more usage intent, and greater leisure usage. While the present study offers only early evidence in this regard, future research can build on these results to extend the CASA paradigm in this direction.

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Appendices

Appendix A. Study 1 Figures



FIGURE 1 Images of e-scooter vandalism or mistreatment taken in the researchers' community.



FIGURE 2 Examples of the base footage messaging and user actions (e.g., the text-only control).



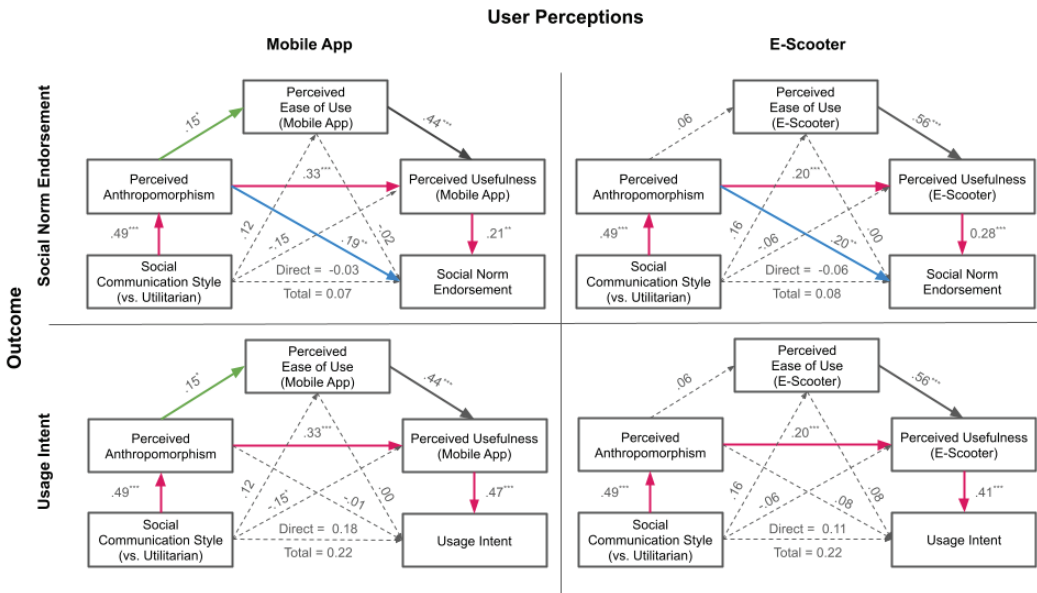
FIGURE 3 E-scooter introductions across the communication-style conditions.



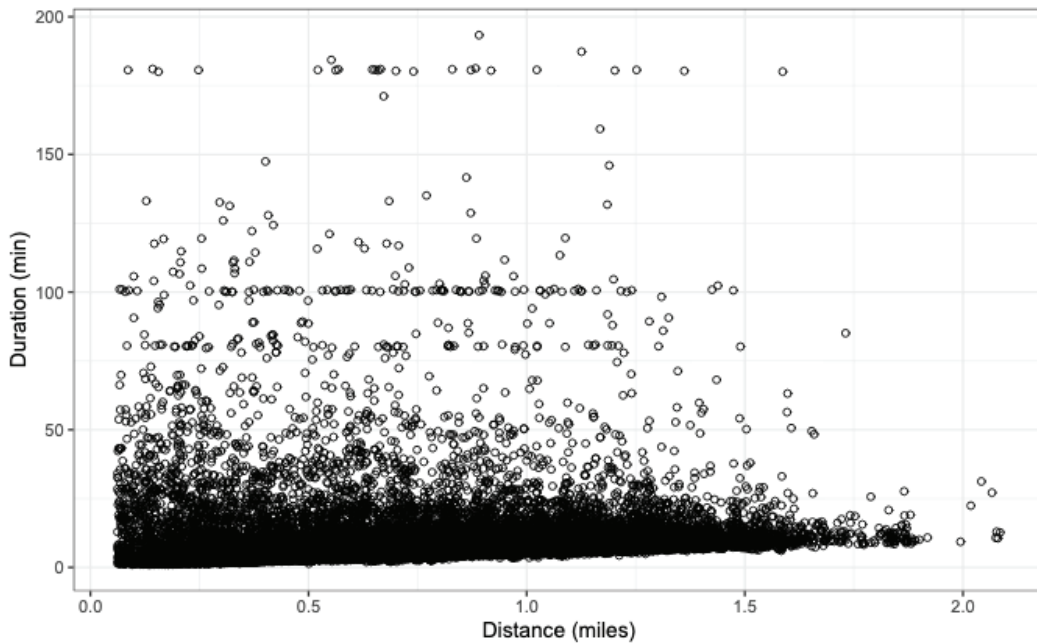
FIGURE 4 All messaging from the female text to speech social condition.

|              | Utilitarian                               | Social Message                                    |
|--------------|-------------------------------------------|---------------------------------------------------|
| Female Voice | Utilitarian-Female Voice<br><i>n</i> = 57 | Social-Female Voice<br><i>n</i> = 54              |
| Male Voice   | Utilitarian-Male Voice<br><i>n</i> = 56   | Social-Male Voice<br><i>n</i> = 55                |
|              |                                           | Text Only<br>(control condition)<br><i>n</i> = 56 |

FIGURE 5 Experimental-design, 2 × 2 factorial design with control group.



**FIGURE 6 Mediation Analysis (Model 6) of four different models.**  
**Note:** The values of paths are standardized coefficients. Significant levels were \*\*\* $p < .05$  \*\* $p < .01$  and \* $p < .001$ . Red arrows indicate the indirect path significant across all four models. Blue arrows indicate the indirect path significant for only models with social norm endorsement as an outcome variable. Green arrows indicate the path significant for models including user perceptions of the mobile app.



**FIGURE 7 Scatter plot between distance and duration of the entire trips.**

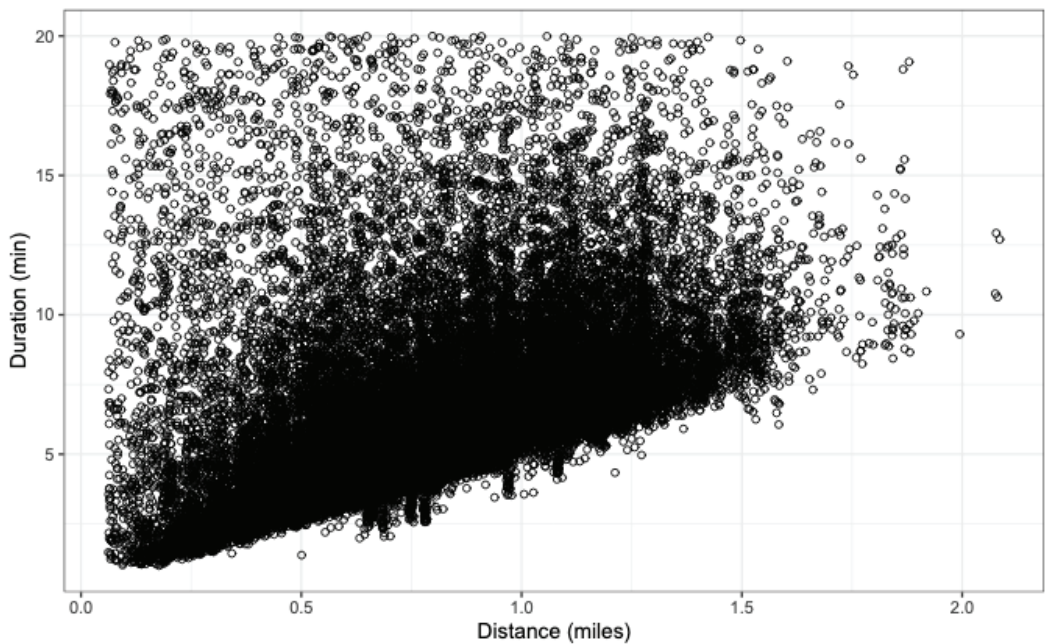


FIGURE 8 Scatter plot between distance and duration of trips below 20 minutes.

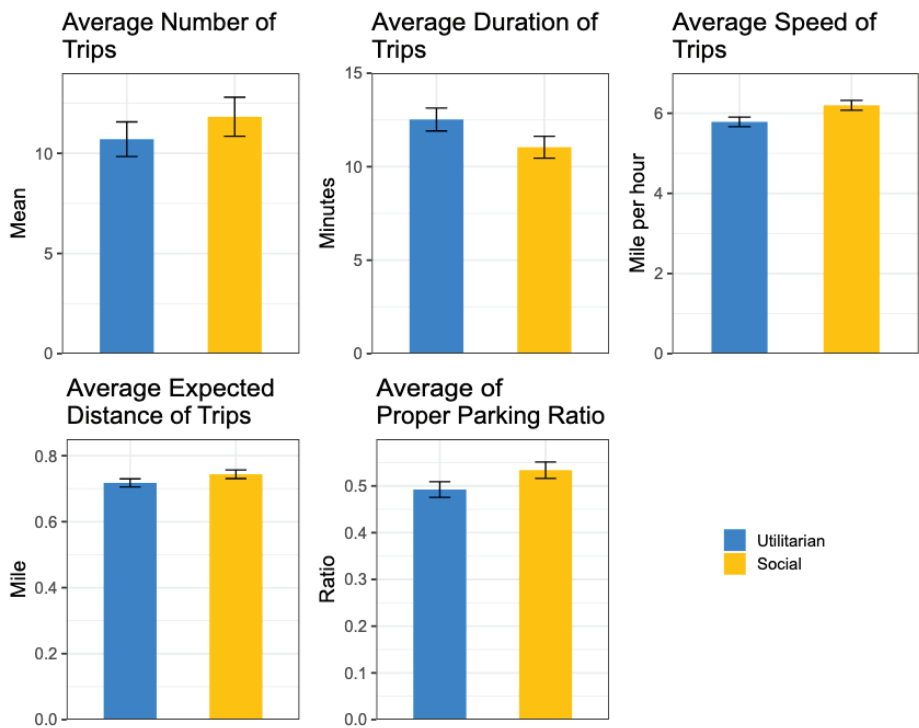
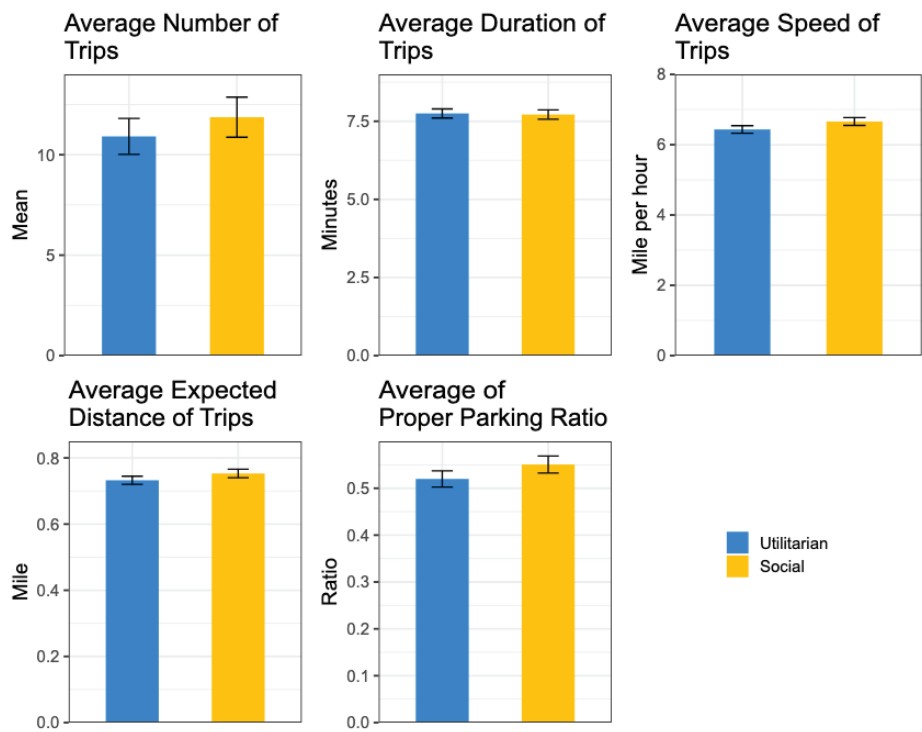


FIGURE 9 Results of Study 2 with the Full Dataset.  
Error bars represent 95% confidence interval.



**FIGURE 10 Results of Study 2 with the Restricted Dataset.**  
Error bars represent 95% confidence interval.

## Appendix B. Tables

**TABLE 1 List of Constructs and Items**

| Measures                          |                                                                         | <i>M</i> | <i>SD</i> | $\alpha$ |
|-----------------------------------|-------------------------------------------------------------------------|----------|-----------|----------|
| <b>Perceived Anthropomorphism</b> |                                                                         | 2.11     | 1.06      | .95      |
|                                   | The e-scooter in the video is able to speak like a human.               |          |           |          |
|                                   | The e-scooter in the video can be happy.                                |          |           |          |
|                                   | The e-scooter in the video can be friendly.                             |          |           |          |
|                                   | The e-scooter in the video can be respectful.                           |          |           |          |
|                                   | The e-scooter in the video can be funny.                                |          |           |          |
|                                   | The e-scooter in the video can be caring.                               |          |           |          |
| <b>Mobile App Usefulness</b>      |                                                                         | 2.95     | 1.01      | .86      |
|                                   | I would find this mobile app useful in my life.                         |          |           |          |
|                                   | Using this mobile app would enable me to accomplish tasks more quickly. |          |           |          |
|                                   | Using this mobile app would increase my productivity.                   |          |           |          |
| <b>Mobile App Ease of Use</b>     |                                                                         | 3.72     | 0.96      | .90      |
|                                   | It would be easy for me to become skillful at using this mobile app.    |          |           |          |
|                                   | I would find this mobile app easy to use.                               |          |           |          |
|                                   | Learning to operate this mobile app would be easy for me.               |          |           |          |
| <b>E-Scooter Usefulness</b>       |                                                                         | 3.41     | 0.88      | .87      |
|                                   | I would find this e-scooter useful in my life.                          |          |           |          |
|                                   | Using this e-scooter would enable me to accomplish tasks more quickly.  |          |           |          |
|                                   | Using this e-scooter would increase my productivity.                    |          |           |          |
| <b>E-Scooter Ease of Use</b>      |                                                                         | 3.88     | 0.87      | .90      |
|                                   | It would be easy for me to become skillful at using this e-scooter.     |          |           |          |
|                                   | I would find this e-scooter easy to use.                                |          |           |          |
|                                   | Learning to operate this e-scooter would be easy for me.                |          |           |          |
| <b>Intention to Use</b>           |                                                                         | 3.30     | 1.02      | .95      |
|                                   | I intend to use them in the future.                                     |          |           |          |
|                                   | I will try to use them in the future.                                   |          |           |          |
|                                   | I expect that I would use them in the future.                           |          |           |          |
|                                   | I plan to use them in the future.                                       |          |           |          |

| Social Norms |                                                                                | 3.19 | 1.01 | .96 |
|--------------|--------------------------------------------------------------------------------|------|------|-----|
|              | I would be embarrassed if the e-scooters were vandalized in my community.      |      |      |     |
|              | If someone in my neighborhood vandalized the e-scooters, I would feel ashamed. |      |      |     |
|              | I would be embarrassed if the e-scooters were damaged in my community.         |      |      |     |
|              | If someone in my neighborhood damaged the e-scooters, I would feel ashamed.    |      |      |     |
|              | I would be embarrassed if the e-scooters were not safely used in my community. |      |      |     |

TABLE 2 Correlations Between Measures

|                        | 1      | 2      | 3      | 4      | 5      | 6      | 7 |
|------------------------|--------|--------|--------|--------|--------|--------|---|
| Anthropomorphism       | –      |        |        |        |        |        |   |
| Mobile App Usefulness  | .349** | –      |        |        |        |        |   |
| Mobile App Ease of Use | .113   | .474** | –      |        |        |        |   |
| E-Scooter Usefulness   | .192** | .626** | .455** | –      |        |        |   |
| E-Scooter Ease of Use  | .019** | .273** | .678** | .526** | –      |        |   |
| Intention to Use       | .72**  | .445** | .212** | .476** | .273** | –      |   |
| Social Norm            | .205** | .300** | .112   | .340** | .131*  | .225** | – |
| * p < .05, ** p < .01  |        |        |        |        |        |        |   |

TABLE 3 Perceived Anthropomorphism, Proto Self-Presence, and Social and Personal Norm Endorsement by E-Scooter Message-Style Condition and Participant Gender

|                        | E-Scooter Messaging-Style Conditions |      |             |      |        |      | Voice Gender |      |        |      |
|------------------------|--------------------------------------|------|-------------|------|--------|------|--------------|------|--------|------|
|                        | Control                              |      | Utilitarian |      | Social |      | Male         |      | Female |      |
|                        | M                                    | SD   | M           | SD   | M      | SD   | M            | SD   | M      | SD   |
| Anthropomorphism       | 1.47                                 | 0.80 | 2.02        | 1.02 | 2.53   | 1.03 | 2.36         | 1.07 | 2.19   | 1.04 |
| Mobile App Usefulness  | 2.60                                 | 0.88 | 3.00        | 1.07 | 3.08   | 0.97 | 3.08         | 1.04 | 3.00   | 1.00 |
| Mobile App Ease of Use | 3.62                                 | 0.91 | 3.66        | 1.02 | 3.83   | 0.92 | 3.76         | 0.98 | 3.73   | 0.97 |
| E-Scooter Usefulness   | 3.31                                 | 0.87 | 3.37        | 0.95 | 3.50   | 0.81 | 3.51         | 0.90 | 3.36   | 0.86 |
| E-Scooter Ease of Use  | 3.89                                 | 0.80 | 3.81        | 0.98 | 3.95   | 0.79 | 3.94         | 0.90 | 3.82   | 0.89 |
| Intention to Use       | 3.29                                 | 1.10 | 3.19        | 1.06 | 3.42   | 0.93 | 3.31         | 1.05 | 3.30   | 0.96 |
| Social Norm            | 3.14                                 | 1.08 | 3.16        | 0.96 | 3.25   | 1.02 | 3.30         | 0.98 | 3.10   | 1.00 |

**TABLE 4** Average Number of Travels, Expected Travel Distances, Trip Duration, Average Speed During the Trip, and Proper Parking Ratio by E-Scooter Message-Style Condition

| Variable                                                                                                                      | Social Message Condition |                   |                  | Utilitarian Message Condition |                   |                  |
|-------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------------|------------------|-------------------------------|-------------------|------------------|
|                                                                                                                               | Mean                     | Median            | SD               | Mean                          | Median            | SD               |
| Number of Trips                                                                                                               | 11.871                   | 5                 | 19.576           | 10.649                        | 4                 | 19.137           |
| Expected Travel Distance*                                                                                                     | 0.712<br>(1145.2)        | 0.720<br>(1159.0) | 0.277<br>(446.1) | 0.678<br>(1090.5)             | 0.702<br>(1128.9) | 0.295<br>(475.2) |
| Trip Duration**                                                                                                               | 11.255                   | 8.061             | 11.921           | 12.792                        | 13.568            | 13.546           |
| Speed of Trip***                                                                                                              | 3.088<br>(4.970)         | 3.090<br>(4.973)  | 0.032<br>(0.051) | 3.086<br>(4.967)              | 3.089<br>(4.971)  | 0.038<br>(0.060) |
| Proper Parking Ratio                                                                                                          | 52.50%                   | 54.50%            | 35.40%           | 48.60%                        | 50%               | 36.90%           |
| Social Message Condition: $n = 1,564$ , Utilitarian Message Condition: $n = 1,938$<br>*Mile (Meters); **Minutes; ***Mph (Kph) |                          |                   |                  |                               |                   |                  |

**TABLE 5** The Results of Levene's Test for Study 2 Variables

| Variables                | Entire Trips |      |          |          | Trips Below 20 Minutes |      |          |          |
|--------------------------|--------------|------|----------|----------|------------------------|------|----------|----------|
|                          | df1          | df2  | <i>F</i> | <i>p</i> | df1                    | df2  | <i>F</i> | <i>p</i> |
| Number of Trips          | 1            | 3500 | 2.22     | .14      | 1                      | 3197 | 1.15     | .28      |
| Expected Travel Distance | 1            | 3500 | 6.66     | .01      | 1                      | 3197 | 1.45     | .23      |
| Trip Duration            | 1            | 3500 | 12.64    | .000     | 1                      | 3197 | 1.49     | .22      |
| Speed of Trip            | 1            | 3500 | 16.61    | .000     | 1                      | 3197 | 4.48     | .03      |
| Proper Parking Ratio     | 1            | 3500 | 7.10     | .01      | 1                      | 3197 | 6.98     | .01      |

## **Appendix C. Study 1 Video Script**

Note: The primary video/footage was consistent between conditions. Only the overlay of the mobile application user interface differed by condition.

### **1. [EXT. LARGE BUILDING—DAY]**

**[ENTIRETY OF FOOTAGE SHOT IN POV of PERSON]**

PERSON walks out of a college building.

PERSON stops and looks at their phone. They notice the time. Frantically, PERSON starts looking around. Suddenly, a parked E-SCOOTER comes into view.

Gleefully, the PERSON walks to the parked E-SCOOTER, pulling their phone back out.

### **2. [EXT. CITY SIDEWALK—DAY]**

PERSON stands next to parked E-SCOOTER, looking at their phone with SPIN UX on screen.

The screen UX is shown next to E-SCOOTER in the air for the viewer (e.g., projection of UX next to E-SCOOTER) as dialogue messages begin to pop up.

**[WIDE SHOT of E-SCOOTER from PERSON POV]**

SPIN UX is transposed into the air next to the E-SCOOTER.

#### **[1.1. FEMALE SOCIAL OTHER CONDITION]**

**SPINIGALE [in robotic FEMALE VO]**

Hi! Spinigale here! I'm close by. Click on the map to see where I am.

#### **[1.2. MALE SOCIAL OTHER CONDITION]**

**SPINNIGAN [in robotic MALE VO]**

Hi! Spinnigan here! I'm close by. Click on the map to see where I am.

#### **[1.3. FEMALE SELF-REPRESENTATION CONDITION]**

**SPIN UX [in robotic FEMALE VO]**

Hi! Your Scooter here. Your scooter is close by. Click on the map to see where your scooter is.

---

**[1.4. MALE SELF-REPRESENTATION CONDITION]****SPIN UX [in robotic MALE VO]**

Hi! Your Scooter here. Your scooter is close by. Click on the map to see where your scooter is.

**[1.5. FEMALE UTILITARIAN CONDITION]****SPIN UX [in robotic FEMALE VO]**

Hi! Scooters are close by. Click on the map to see where they are.

**[1.6. MALE UTILITARIAN CONDITION]****SPIN UX [in robotic MALE VO]**

Hi! Scooters are close by. Click on the map to see where they are.

PERSON taps RENT.

**[2.1. FEMALE SOCIAL OTHER CONDITION]****SPINIGALE [in robotic FEMALE VO]**

Hi! Spinigale here. Thanks for renting me! Let's have a safe and fun ride!

**[2.2. MALE SOCIAL OTHER CONDITION]****SPINNIGAN [in robotic MALE VO]**

Hi! Spinnigan here. Thanks for renting me! Let's have a safe and fun ride!

**[2.3. FEMALE SELF-REPRESENTATION CONDITION]****SPIN UX [in robotic FEMALE VO]**

Hi! Your Scooter here. Thanks for renting! Hope you have a safe and fun ride!

**[2.4. MALE SELF-REPRESENTATION CONDITION]****SPIN UX [in robotic MALE VO]**

Hi! Your Scooter here. Thanks for renting! Hope you have a safe and fun ride!

**[2.5. FEMALE UTILITARIAN CONDITION]****SPIN UX [in robotic FEMALE VO]**

Hi! Thanks for renting this scooter! Have a safe and fun ride!

**[2.6. MALE UTILITARIAN CONDITION]****SPIN UX [in robotic MALE VO]**

Hi! Thanks for renting this scooter! Have a safe and fun ride!

UX disappears as the person puts the phone away and walks to get on E-SCOOTER. They start their ride and zoom away.

---

### **3. [EXT. ROAD—DAY]**

[POV as PERSON rides E-SCOOTER on the road, going quite fast]

UX appears again: PERSON notices and looks down.

**[3.1. FEMALE SOCIAL OTHER CONDITION]**

**SPINIGALE [in robotic FEMALE VO]**

Hey, Spinigale again. Please slow me down! This is a slow zone.

**[3.2. MALE SOCIAL OTHER CONDITION]**

**SPINNIGAN [in robotic MALE VO]**

Hey, Spinnigan again. Please slow me down! This is a slow zone.

**[3.3. FEMALE SELF-REPRESENTATION CONDITION]**

**SPIN UX [in robotic FEMALE VO]**

Please slow Your Scooter down! This is a slow zone.

**[3.4. MALE SELF-REPRESENTATION CONDITION]**

**SPIN UX [in robotic MALE VO]**

Please slow Your Scooter down! This is a slow zone.

**[3.5. FEMALE UTILITARIAN CONDITION]**

**SPIN UX [in robotic FEMALE VO]**

Please slow the scooter down! This is a slow zone.

**[3.6. MALE UTILITARIAN CONDITION]**

**SPIN UX [in robotic MALE VO]**

Please slow the scooter down! This is a slow zone.

PERSON looks around embarrassingly and slows down. UX disappears, and PERSON and E-SCOOTER zoom off.

### **4. [EXT. BUILDING—DAY]**

PERSON and E-SCOOTER pull in front of a building. The UX overlay appears again, and it is tracked onto E-SCOOTER as it slows.

**[4.1. FEMALE SOCIAL OTHER CONDITION]**

**SPINIGALE [in robotic FEMALE VO]**

It's me, Spinigale! Thanks for a great ride! Please park me somewhere safe.

---

**[4.2. MALE SOCIAL OTHER CONDITION]**

**SPINNIGAN [in robotic MALE VO]**

It's me, Spinnigan! Thanks for a great ride! Please park me somewhere safe.

**[4.3. FEMALE SELF-REPRESENTATION CONDITION]**

**SPIN UX [in robotic FEMALE VO]**

Thanks for a great ride! Please park Your Scooter somewhere safe.

**[4.4. MALE SELF-REPRESENTATION CONDITION]**

**SPIN UX [in robotic MALE VO]**

Thanks for a great ride! Please park Your Scooter somewhere safe.

**[4.5. FEMALE UTILITARIAN CONDITION]**

**SPIN UX [in robotic FEMALE VO]**

Thanks for a great ride! Please park this scooter somewhere safe.

**[4.6. MALE UTILITARIAN CONDITION]**

**SPIN UX [in robotic MALE VO]**

Thanks for a great ride! Please park this scooter somewhere safe.

**VANDALISM MANIPULATION**

**[1 SOCIAL OTHER CONDITION]**

Thank you for your safe and friendly treatment of me and for encouraging others to do the same!

**[2 SELF-REPRESENTATION CONDITION]**

Thank you for your safe and friendly treatment of Your Scooter and for encouraging others to do the same!

**[3 UTILITARIAN CONDITION]**

Thank you for your safe and friendly treatment of this scooter and for encouraging others to do the same!

They slow down, and PERSON gets off the E-SCOOTER, parking it in the parking space. PERSON pulls out their phone. PERSON takes a picture of the parked E-SCOOTER.

**5. [POV WIDE-SHOT of E-SCOOTER]**

**[5.1. FEMALE SOCIAL OTHER CONDITION]**

**SPINIGALE [in robotic FEMALE VO]**

Spinigale here. Thanks! Hope you take me for a ride again sometime!

**[5.2. MALE SOCIAL OTHER CONDITION]**

**SPINNIGAN [in robotic MALE VO]**

Spinnigan here. Thanks! Hope you take me for a ride again sometime!

**[5.3. FEMALE SELF-REPRESENTATION CONDITION]**

**SPIN UX [in robotic FEMALE VO]**

Thanks! Hope you take Your Scooter for a ride again sometime!

**[5.4. MALE SELF-REPRESENTATION CONDITION]**

**SPIN UX [in robotic MALE VO]**

Thanks! Hope you take Your Scooter for a ride again sometime!

**[5.5. FEMALE UTILITARIAN CONDITION]**

**SPIN UX [in robotic FEMALE VO]**

Thanks! Hope you take a scooter for a ride again sometime!

**[5.6. MALE UTILITARIAN CONDITION]**

**SPIN UX [in robotic MALE VO]**

Thanks! Hope you take a scooter for a ride again sometime!

PERSON puts the phone in pocket and walks away. UX overlay shows a five-star ride rating and then disappears as they turn away.

**[FADE OUT]**

**[END]**

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